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Effects of the CO/CO_2 Ratio in Synthesis Gas on the Catalytic Behavior in Fischer−Tropsch Synthesis Using K/Fe−Cu−Al Catalysts

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ABSTRACT: The catalytic behavior of a K/Fe−Cu−Al catalyst has been investigated together with carbon deposition by variation of the $CO_2/(CO + CO_2)$ ratios in H_2-sufficient feed gas during the Fischer−Tropsch synthesis (FTS). Both fresh and used catalysts were characterized by means of X-ray diffraction, temperature-programmed decarburization, thermogravimetric analysis, and elemental analysis. Under H_2-sufficient and CO_2-abundant conditions [$H_2/(2CO + 3CO_2) = 1$, and $CO_2/(CO + CO_2) \geq 0.5$], CO_2 in the feed gas was consumed for hydrocarbon production. The CO_x conversion to hydrocarbons increased with the temperature, pressure, and CO content in the feed gas. Under high-pressure reaction conditions, carbon deposition was significantly reduced, as compared to that at low pressures. Overall, conditions involving a high $CO_2/(CO + CO_2)$ ratio, temperature, and pressure are beneficial for both CO_2 utilization and prevention of carbon deposition in the FTS using Fe catalysts.

1. INTRODUCTION

Recently, there has been an increase in the concern over the effects of global warming. Particularly, research interest directed toward CO_2 fixation and conversion for mitigating greenhouse effects has increased over the past decade, because CO_2 is a typical greenhouse gas arising from human activities (e.g., in automobiles, electricity generation, and industrial processes).[1−3] Various methods have been suggested for reducing the concentration of atmospheric CO_2 or its recycling. Among these suggestions, the catalytic conversion of CO_2 has received much attention as a promising technique for CO_2 utilization.[4−6] Thus, as one way for the utilization of CO_2, many groups have investigated the catalytic conversion of CO_2 using methods such as gas-to-liquid (GTL) processes, methanol synthesis, and dimethyl ether (DME) synthesis.[7−10]

The GTL process has received attention as an alternative method for supplying hydrocarbons (HCs), owing to the sharp increase in petroleum prices over the past decade, and possesses a number of advantages: first, a high value-added product can be obtained from natural gas; second, product handling and transportation are easy because of its liquid properties; and third, a clean liquid fuel, which does not include sulfur and aromatic compounds, can be supplied.[11,12] Typically, Co catalysts have been used in the GTL process because of their high activities and longer lifetimes. There have been intensive efforts directed at the development of GTL processes using CO_2 as a co-feed, which employs CO_2/steam-mixed reforming for synthesis gas generation and Co catalysts for the Fischer−Tropsch synthesis (FTS).[13,14] However, excessive CO_2 feeding in the reforming process is necessary to give a synthesis gas composition with $H_2/CO = 2$.[15,16] In FTS routes using a Co catalyst, the unconverted CO_2 in the reforming process acts as an inert gas because the Co species shows almost no activity in the water−gas shift (WGS) reaction. Therefore, the unconverted CO_2 should be separated and recycled back to the reforming process or re-emitted back to air. However, when considering the use of an Fe catalyst in the GTL process, carbon efficiency is expected to improve because the CO_2 utilization would be enhanced as a result of CO_2 conversion in the FTS process.

Because the FTS as a part of the GTL process is highly exothermic, it is important to rapidly remove the heat of reaction to avoid high production of CH_4. In the case of Fe catalysts, CH_4 selectivity is not significantly increased even in reactions at high temperatures. Also, gasoline and linear low-molecular-mass olefins can be produced in high yields. Above all, the high rate of CO_2 utilization is feasible, because Fe catalysts demonstrate high activity for both FTS and WGS reactions.[17−19] In the FTS reaction, CO_2 hydrogenation progresses in two steps: (1) CO production because of the reverse WGS (RWGS) reaction and (2) hydrogenation by the general FTS reaction. In our previous work, Fe catalysts prepared by co-precipitation showed high performances toward HC synthesis from CO_2-containing feed gas under H_2-sufficient conditions.[20,21] Also, Matsumoto et al.[22] reported that the FTS reaction performed under H_2-insufficient conditions resulted in increased carbon deposition, as compared to that performed under H_2-sufficient conditions. In principle, the Fe catalyst should be an ideal candidate for use in the FTS process using CO_2-containing feed gas. However, carbon deposition, which induces catalytic deactivation because of pore plugging and catalyst breaking, is a fundamentally significant problem encountered in commercial processes employing Fe catalysts.[23,24] Many groups have investigated the effect of CO/CO_2 feed gas ratios on the FTS reaction. However, only reports of catalytic performance exist in the literature; to the best of our knowledge, no studies providing details of carbon deposition

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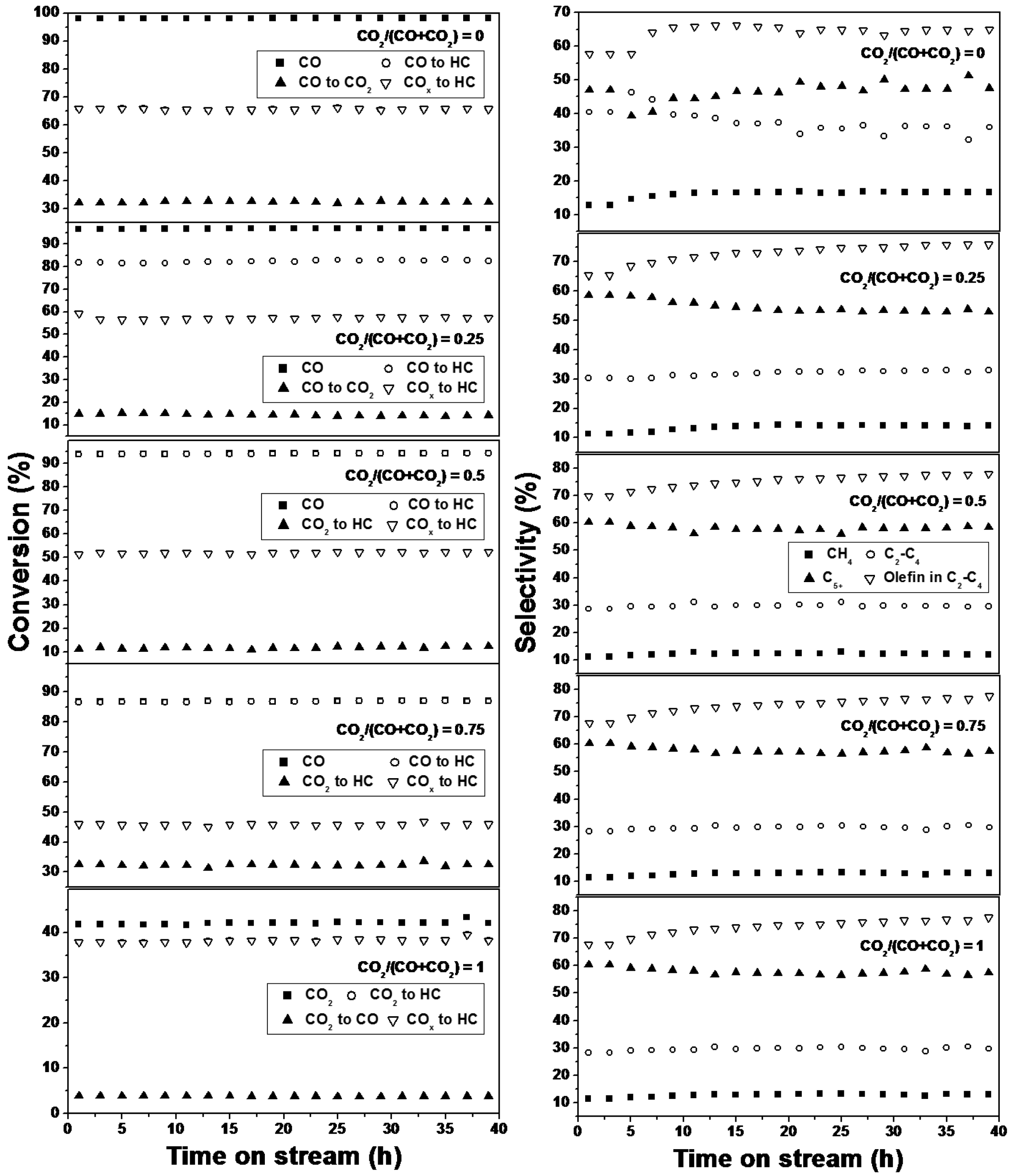

Figure 1. Changes in conversion and selectivity with varying $CO_2/(CO + CO_2)$ ratios in FTS over time on-stream at 320 °C and 10 bar.

on Fe catalysts exist, despite its importance for catalyst physical strength.[5,25]

In this study, we discuss the effect of reaction conditions on carbon formation on the Fe catalyst during the FTS process. For this, the FTS reactions using a K/Fe−Cu−Al catalyst were performed with varying temperatures, pressures, and $CO_2/(CO + CO_2)$ ratios in H_2-sufficient feed gas [$H_2/(2CO + 3CO_2) = 1$]. After reaction, the amount of deposited carbon on the catalyst was investigated through results of analysis.

2. EXPERIMENTAL SECTION

2.1. Catalyst Preparation. The K/Fe−Cu−Al catalyst was prepared by the co-precipitation of Fe−Cu−Al precursors, followed by potassium impregnation. The Fe−Cu−Al precursors were prepared by continuous co-precipitation from an aqueous solution of metal nitrates with ammonium hydroxide at a constant pH of 7.0. The resulting precipitate was subjected to filtrations and washing cycles, then dried in air at 110 °C overnight, and calcined at 450 °C for 5 h. The obtained product was impregnated with potassium carbonate using the incipient wetness technique to prepare the K/Fe−Cu−Al sample, followed by drying and calcination under the same conditions. The catalyst was sieved to the required particle size (2−2.4 mm). The

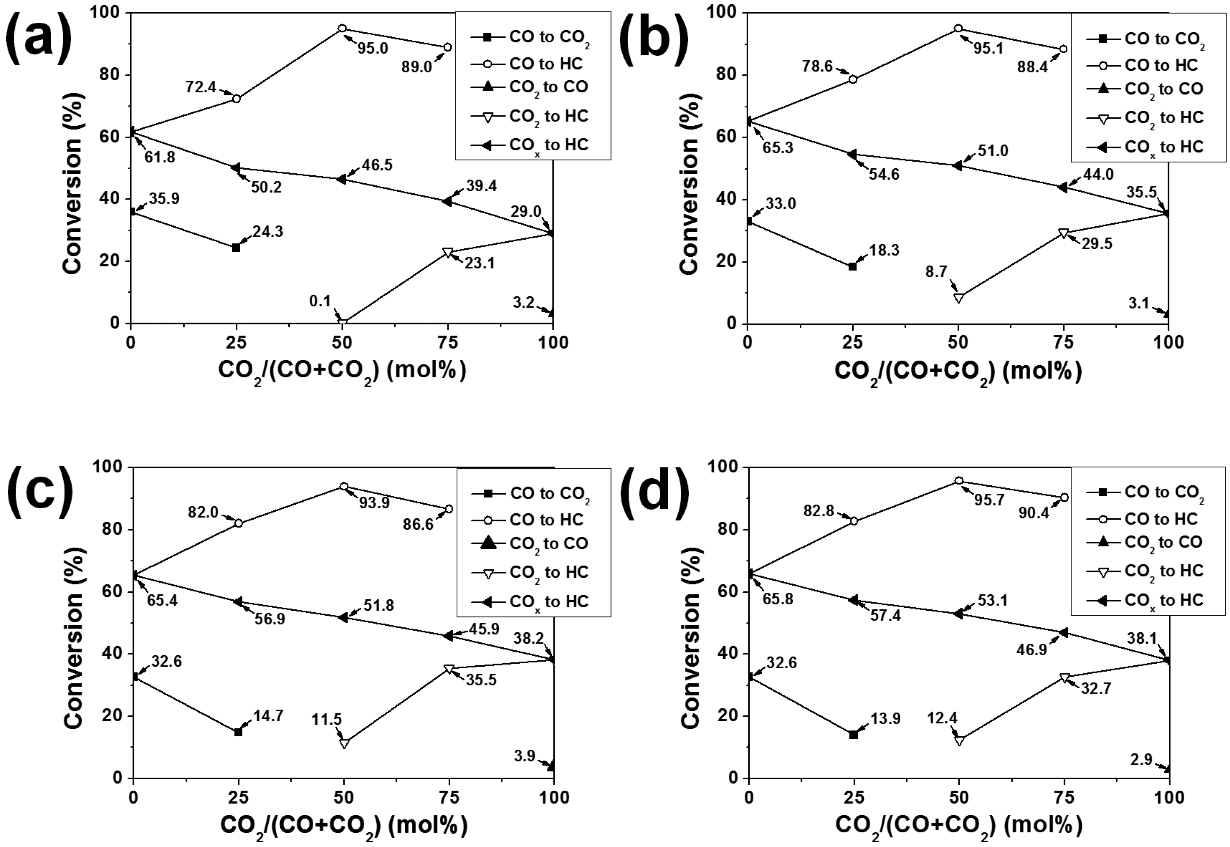

Figure 2. Variation in conversions depending upon $CO_2/(CO + CO_2)$ ratios at (a) 280 °C and 10 bar, (b) 300 °C and 10 bar, (c) 320 °C and 10 bar, and (d) 300 °C and 20 bar.

final composition (in mass ratio) of the catalyst was 4K/100Fe-6Cu-16Al. With regard to catalyst composition, high potassium content is beneficial for CO_2 hydrogenation and the addition of copper is known to facilitate the reduction of iron. Alumina addition is suggested to improve the iron dispersion of the catalyst.[17,26,27] In our earlier study,[21] the performance of K/Fe−Cu−Al in the FTS was investigated for varying potassium contents. On the basis of these results, a potassium content of K/Fe = 0.04 was selected as the optimum level in the present study.

2.2. Catalyst Characterization. The characterization of used catalysts was carried out by means of X-ray diffraction (XRD), temperature-programmed decarburization (TPDC), thermogravimetric analysis (TGA), and elemental analysis. All of the catalysts were washed with cyclohexane to remove deposited HCs prior to analysis. The freshly reduced catalyst was also characterized by TGA, TPDC, and elemental analysis to determine the amount of carbide formation after passivating it with 0.1% O_2/He gas at room temperature for 1 h. XRD patterns were obtained at room temperature in the range 2θ = 5−80° using a Rigaku D/Max-RC diffractometer equipped with a graphite (002) crystal as a source of monochromatic Cu Kα radiation. TPDC profiles were obtained while raising the temperature of the catalyst bed under a H_2 flow of 20 mL/min to 900 °C at a heating rate of 10 °C/min. Analysis of the produced carbides was carried out by converting them to CH_4 prior to their introduction into a flame ionization detector (FID). TGA of catalysts after reaction was carried out using a TA Instrument SDT Q600 V20.5 Build. The samples were heated under an air flow of 100 mL/min to 1000 °C at a heating rate of 10 °C/min. Elemental analysis of the catalyst after reaction was carried out using a Thermo Scientific Flash 2000 organic elemental analyzer.[28]

2.3. Catalytic Reaction. Two types of gas mixtures consisting of $CO/H_2/Ar$ (31.5:63:5.5 molar ratio, denoted as gas mixture 1) and $CO_2/H_2/Ar$ (24:72:4 molar ratio, denoted as gas mixture 2) were used as a feed gas for the FTS reaction. The catalysts were reduced *in situ* using gas mixture 1 for 2 h at 300 °C under 10 bar, followed by cooling to 50 °C. The feed gas was balanced to a ratio of $H_2/(2CO + 3CO_2)$ = 1 by mixing gas mixtures 1 and 2. The FTS reactions were performed using the balanced feed gas with varying $CO_2/(CO + CO_2)$ ratios (0, 0.25, 0.5, 0.75, and 1) in a fixed-bed reactor under 10 bar and varying temperatures (280, 300, and 320 °C) with a space velocity of 1800 [normal temperature and pressure (NTP)] mL g^{-1} of catalyst h^{-1}. The FTS reaction was also performed at 300 °C under 20 bar to investigate the effect of the reaction pressure. The effluent from the reactor was periodically injected into online gas chromatography (GC) using one six-port valve. Ar was used as an internal standard to calculate product selectivity of CO conversions. Ar, CO_2, CO, and CH_4 were analyzed with a thermal conductivity detector (TCD) equipped with a packed column (Carboxen 1000, Supelco), and HCs (C_1−C_4 and C_{5+}) were analyzed with a FID equipped with a capillary column (PLOT/Q, HP).

3. RESULTS AND DISCUSSION

3.1. Catalytic Conversion and Selectivity. FTS was carried out using varying $CO_2/(CO + CO_2)$ ratios and temperatures to investigate the effects of reaction conditions on catalytic conversion and selectivity. Because catalytic conversions and selectivities over time on-stream showed almost the same pattern under all conditions, the patterns at 320 °C under 10 bar are presented as a representative result in Figure 1. The results obtained under the other conditions are presented in Figures 2 and 3. Catalytic conversions showed no significant decrease over 40 h on-stream. CO conversions were maintained over 95%, and CO was also converted to CO_2 via the WGS reaction at $CO_2/(CO + CO_2)$ ratios from 0 to 0.25. At $CO_2/(CO + CO_2)$ ratios from 0.5 to 0.75, CO conversion was decreased to less than 95% because of the generation of

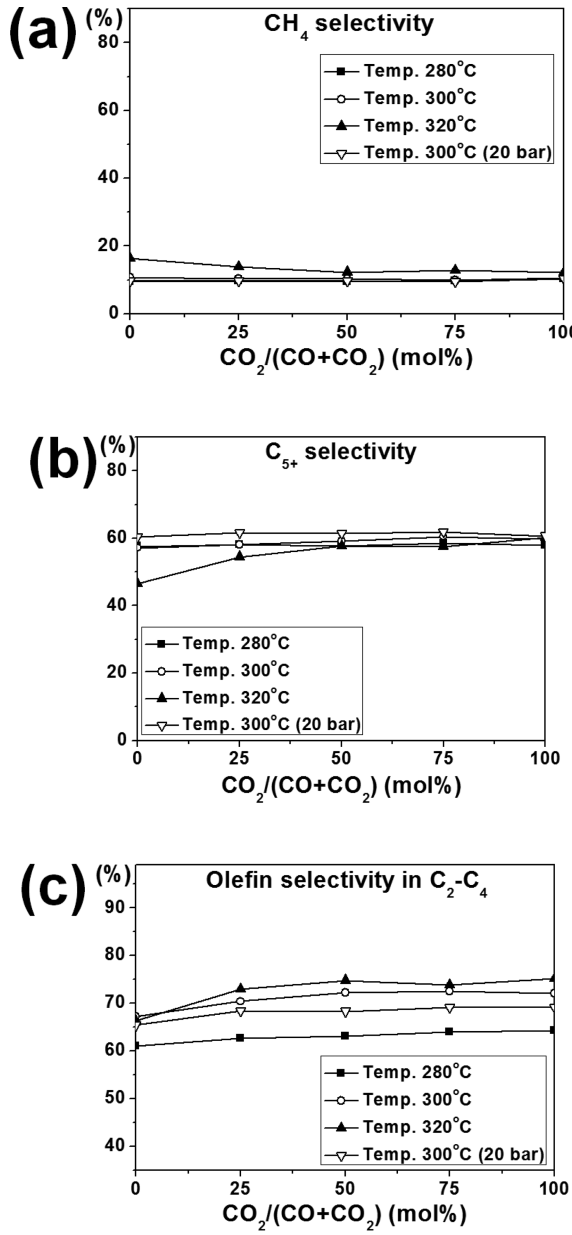

Figure 3. Variations in selectivity depending upon $CO_2/(CO + CO_2)$ ratios: (a) CH_4 selectivity, (b) C_{5+} selectivity, and (c) C_2-C_4 fraction olefin selectivity.

CO by the RWGS reaction.[29] CH_4 and C_2-C_4 olefin selectivities were gradually increased over 40 h on-stream. CH_4 selectivities showed low values of around 10% under all conditions, whereas olefin selectivities in the C_2-C_4 fraction showed high values of around 65% under all conditions. Alkali metals as catalyst components are known to cause a drastic decline in CH_4 selectivity, accompanied by a strong increase in the olefinic preference of the C_2-C_4 HCs.[26,27] In contrast, C_{5+} selectivities were decreased during 40 h on-stream. It can be inferred that the CH_4 and C_{5+} selectivities competitively affect each other.

Figure 2 shows the variation in conversions depending upon $CO_2/(CO + CO_2)$ ratios under various reaction conditions. The CO conversion to HC showed the highest value at $CO_2/(CO + CO_2) = 0.5$ and then slightly decreased at $CO_2/(CO + CO_2) = 0.75$. As presented in Figure 2a, particularly, CO_2 conversion to HC was nearly 0% at $CO_2/(CO + CO_2) = 0.5$ at 280 °C because the WGS reaction was nearly equilibrated at this condition.[25,30] In most cases, CO_2 was converted to HCs at $CO_2/(CO + CO_2)$ ratios over 0.5, owing to the RWGS reaction besides the reaction at 280 °C, and its conversion increased with the $CO_2/(CO + CO_2)$ ratio.[29] CO_x conversion to HC increased with the reaction temperature and CO content in the feed gas. With an increase in the reaction pressure to 20 bar at 300 °C, CO_x conversion to HC was increased and showed the highest value among those observed under all conditions. It can be inferred that the increase in CO_x conversion to HCs is due to the increased density of adsorbed CO and H_2 molecules on the catalyst surface under higher pressure. The CO_x conversion to HC decreased with an increasing $CO_2/(CO + CO_2)$ ratio because of the superiority of the RWGS reaction. Moreover, strongly oxidizing H_2O generated during the FTS reaction under iron catalysis acts as an inhibitor, thereby limiting high CO_x conversion per pass. This effect grows in increasing order of the $CO_2/(CO + CO_2)$ ratio, in the order of their increasing RWGS reaction rates.[20,31] Also, the lower partial pressure of CO on the catalyst surface makes the CO_x conversion lower because of the carbiding potential decrease.[20,32]

Figure 3 shows the variation in selectivity depending upon the $CO_2/(CO + CO_2)$ ratios under various reaction conditions. As shown in Figure 3a, no significant change in CH_4 selectivity could be observed, with the exception of reactions performed at 320 °C under 10 bar, at which CH_4 selectivities were higher and decreased with increasing $CO_2/(CO + CO_2)$ ratios. In most cases, CH_4 selectivity did not increase with the $CO_2/(CO + CO_2)$ ratio; in fact, a slight decrease was observed at 320 °C. In our previous work, particularly, CH_4 selectivity remained low when the synthesis gas contained only CO_2 and no CO, indicating this unique characteristic of the iron catalyst.[20,32] The selectivities of C_{5+} HCs in Figure 3b showed no significant change with the $CO_2/(CO + CO_2)$ ratio at 280 and 300 °C. The selectivities of C_{5+} HCs showed the highest value at 300 °C under 20 bar, a result attributed to the promotion of chain propagation because of the increase in surface carbon oxide molecules generated by high pressure.[19] In the case of results obtained at 320 °C, C_{5+} selectivity increased with greater $CO_2/(CO + CO_2)$ ratios at the expense of CH_4 selectivity. This can be explained in terms of water effects on the selectivity. As shown in Figure 2, an increase in the reaction temperature resulted in a decrease in CO conversion to CO_2 and an increase in CO_2 conversion to HC. The rate of the RWGS reaction is enhanced at increasing temperatures and $CO_2/(CO + CO_2)$ ratios. Thus, the concentration of surface H_2O is increased because of the elevation of CO_2 and H_2 consumption, resulting from enhanced RWGS reactivity. Satterfield et al.[33] reported that CH_4 selectivity decreased with increasing H_2O feed because of the inhibition of hydrogenation by H_2O. For the same reason, the results obtained at 320 °C under 10 bar showed a decrease in CH_4 and an increase in C_{5+} selectivity with an increasing $CO_2/(CO + CO_2)$ ratio.

Figure 4 shows the rate of H_2O formation during the FTS reaction depending upon the $CO_2/(CO + CO_2)$ ratio at varying temperatures. The rate of H_2O formation increased with the temperature and $CO_2/(CO + CO_2)$ ratio. It can thus be inferred that the rate of the RWGS reaction is increased at elevated temperatures and $CO_2/(CO + CO_2)$ ratios. Furthermore, this result supports the effect of generated H_2O as a catalyst oxidizer and hydrogenation inhibitor during the FTS process.

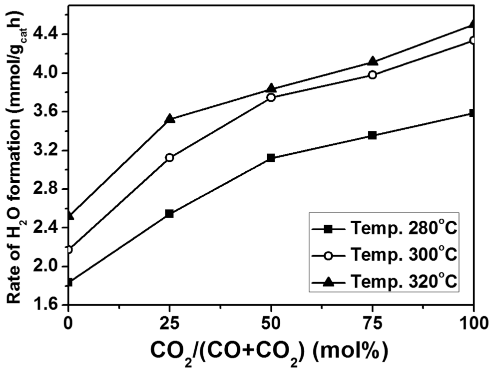

Figure 4. Rate of H_2O formation during the FTS reaction depending upon CO_2/(CO + CO_2) ratios at varying temperatures.

3.2. Equilibrium Calculation of the WGS Reaction.

Figure 5 shows the temperature-dependent equilibrium conversion of the WGS reaction at varying CO_2/(CO + CO_2) ratios. In Figure 5a, all calculated CO conversions showed zero or negative values, an expected outcome because only CO formation occurs in the absence of H_2O when taking only WGS into consideration, and decreases with increasing temperature. However, the calculated CO conversions decreased with increasing CO_2/(CO + CO_2) ratios and matched experimental data obtained at high CO_2/(CO + CO_2) ratios. In comparison to the experimental results, the calculated CO conversion showed a significant discrepancy at low CO_2/(CO + CO_2) ratios. This discrepancy is mainly due to H_2O formation during FTS. While H_2O was not considered as a feed component in the thermodynamic calculation, it was formed during FTS in the actual reaction. Teng et al. also investigated the temperature dependence of the WGS reaction using Fe−Mn catalysts.[34] According to their investigation, the RWGS reaction rate was increased with the temperature; however, the calculated data were not exactly matched with experimental data because of the interference of H_2O formed by FTS. The calculated conversion of CO_2 in Figure 5b

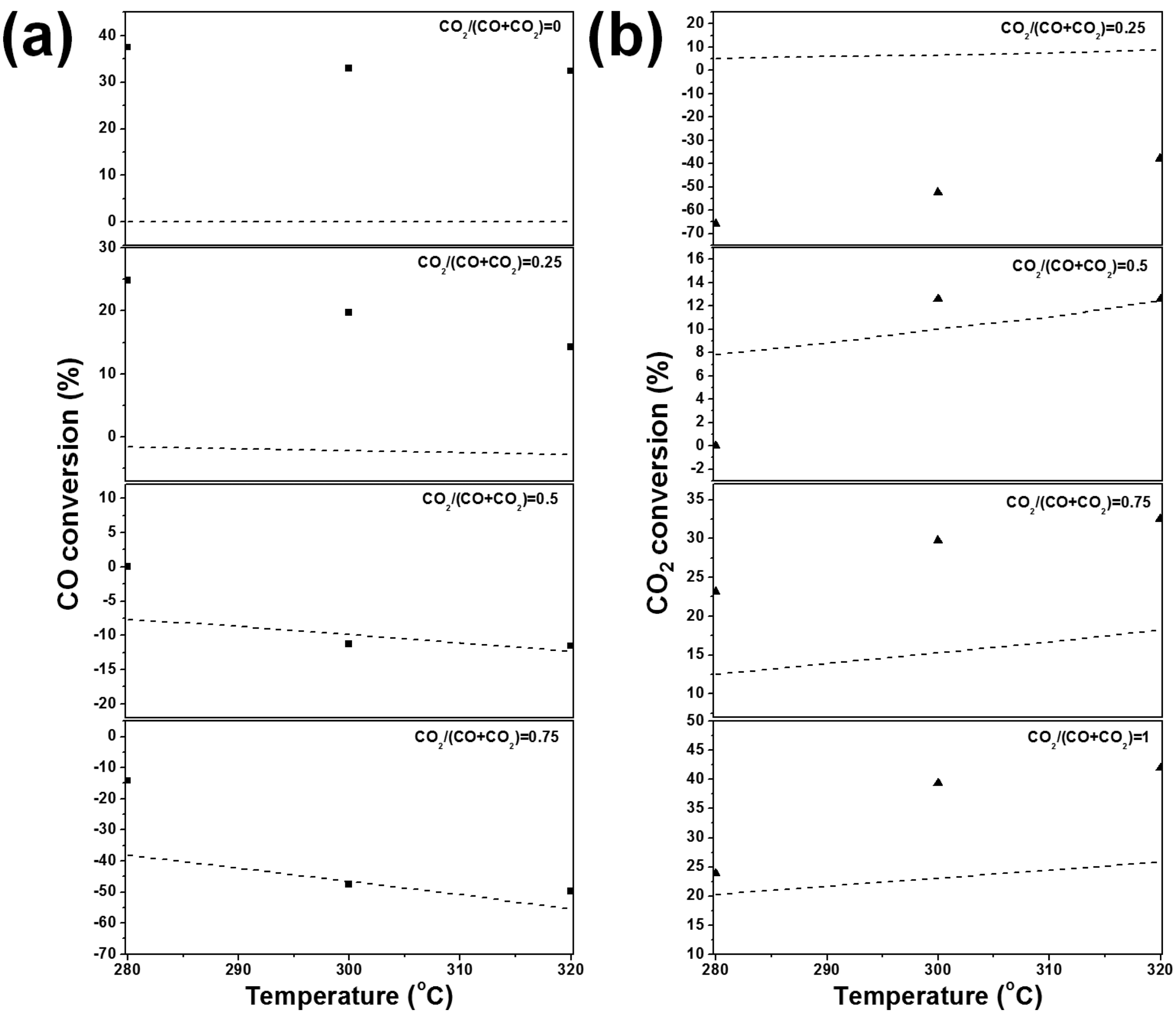

Figure 5. Temperature-dependent equilibrium conversion of the WGS reaction and experimental data at varying CO_2/(CO + CO_2) ratios: (- - -) calculated conversion of WGS, (■) experimental CO conversion to CO_2, and (▲) experimental total conversion of CO_2.

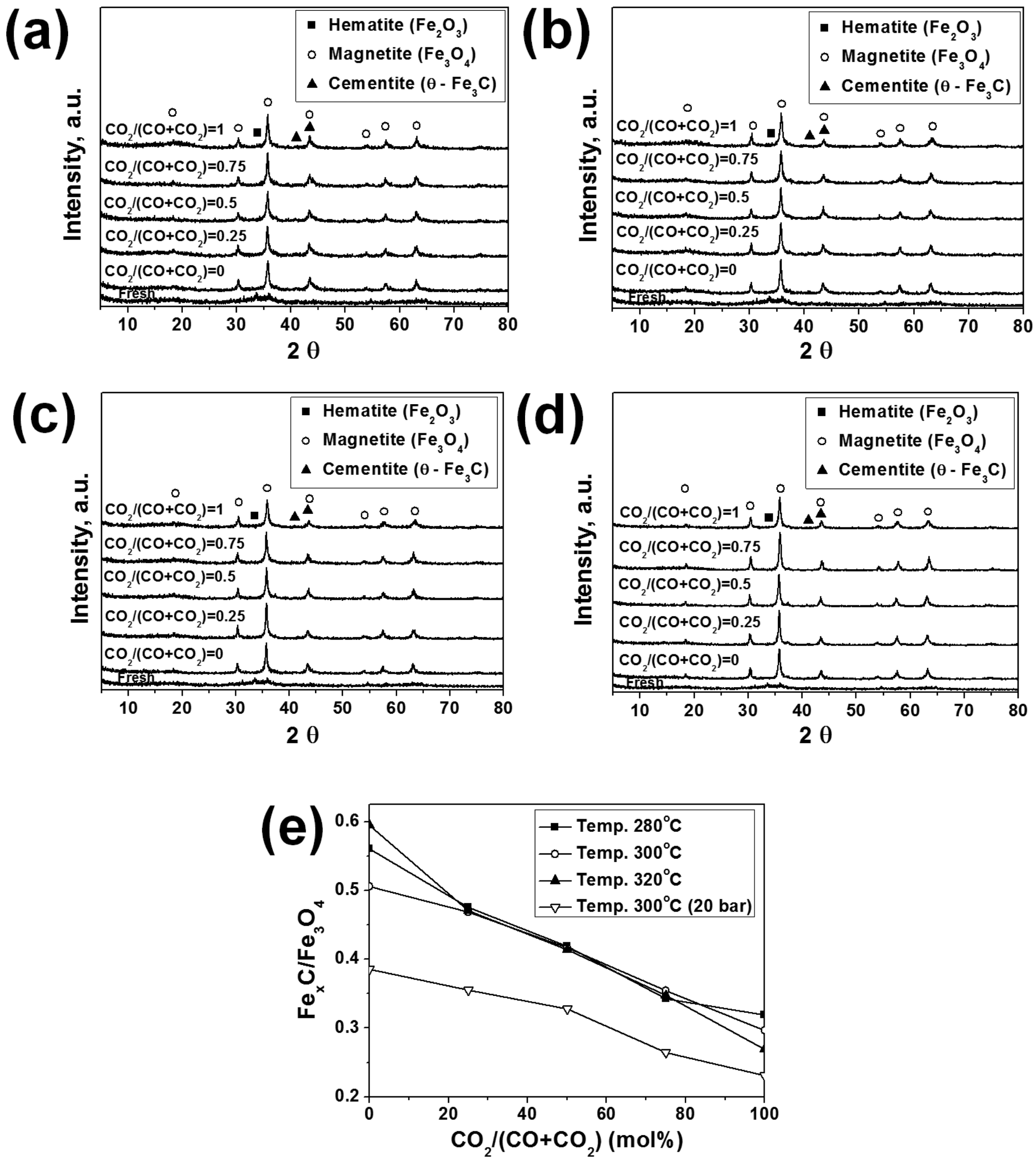

Figure 6. XRD patterns of fresh and used catalyst at (a) 280 °C and 10 bar, (b) 300 °C and 10 bar, (c) 320 °C and 10 bar, and (d) 300 °C and 20 bar. (e) Fe_xC/Fe_3O_4 ratio estimated from XRD patterns.

increased with the temperature and $CO_2/(CO + CO_2)$ ratio, a result that is in good agreement with the previous observations made by other investigators regarding the WGS reaction.[35,36] However, the calculated CO_2 conversion also showed a significant discrepancy with experimental data. Likewise, this result is due to the generated H_2O, which favors the forward WGS reaction. Furthermore, experimental data at high $CO_2/(CO + CO_2)$ ratios showed higher CO_2 conversions than calculated data, indicating that the WGS equilibrium shifts to the reverse direction as a result of CO consumption by FTS.

3.3. XRD Analysis. Panels a–d of Figure 6 show the XRD patterns of fresh and used catalysts at varying $CO_2/(CO + CO_2)$ ratios at each condition examined. No significant difference among the XRD patterns of the used catalyst could be observed under each set of conditions. Two broad peaks were observed in the XRD pattern of the fresh catalyst, which can be attributed to the presence of hematite [Joint Committee on Powder Diffraction Standards (JCPDS) card number 33-0664] and magnetite (JCPDS card number 39-1346) at 2θ = 33.5° and 36.0°, respectively.[37,38] The intensity of magnetite (Fe_3O_4) peaks between 18.0° and 55.0° was significantly increased because of oxidation and sintering during the reaction. The XRD patterns of the used catalysts showed the appearance of peaks at 2θ = 40.7° and 43.0° corresponding to cementite (JCPDS card number 39-1346).[39,40] The iron carbide peak intensities for used catalysts decreased with increasing $CO_2/(CO + CO_2)$ ratios.

Figure 6e shows the patterns of Fe_xC/Fe_3O_4 ratios calculated by integrating the highest metal oxide (2θ = 36.0°) and carbide (2θ = 43.0°) peaks in the XRD patterns. The Fe_xC/Fe_3O_4 ratio was observed to decrease as the $CO_2/(CO + CO_2)$ ratio was increased. The Fe_xC/Fe_3O_4 ratio of the sample used at 300 °C under 20 bar was significantly lower than those used under the other conditions. On the basis of the XRD results, it can be

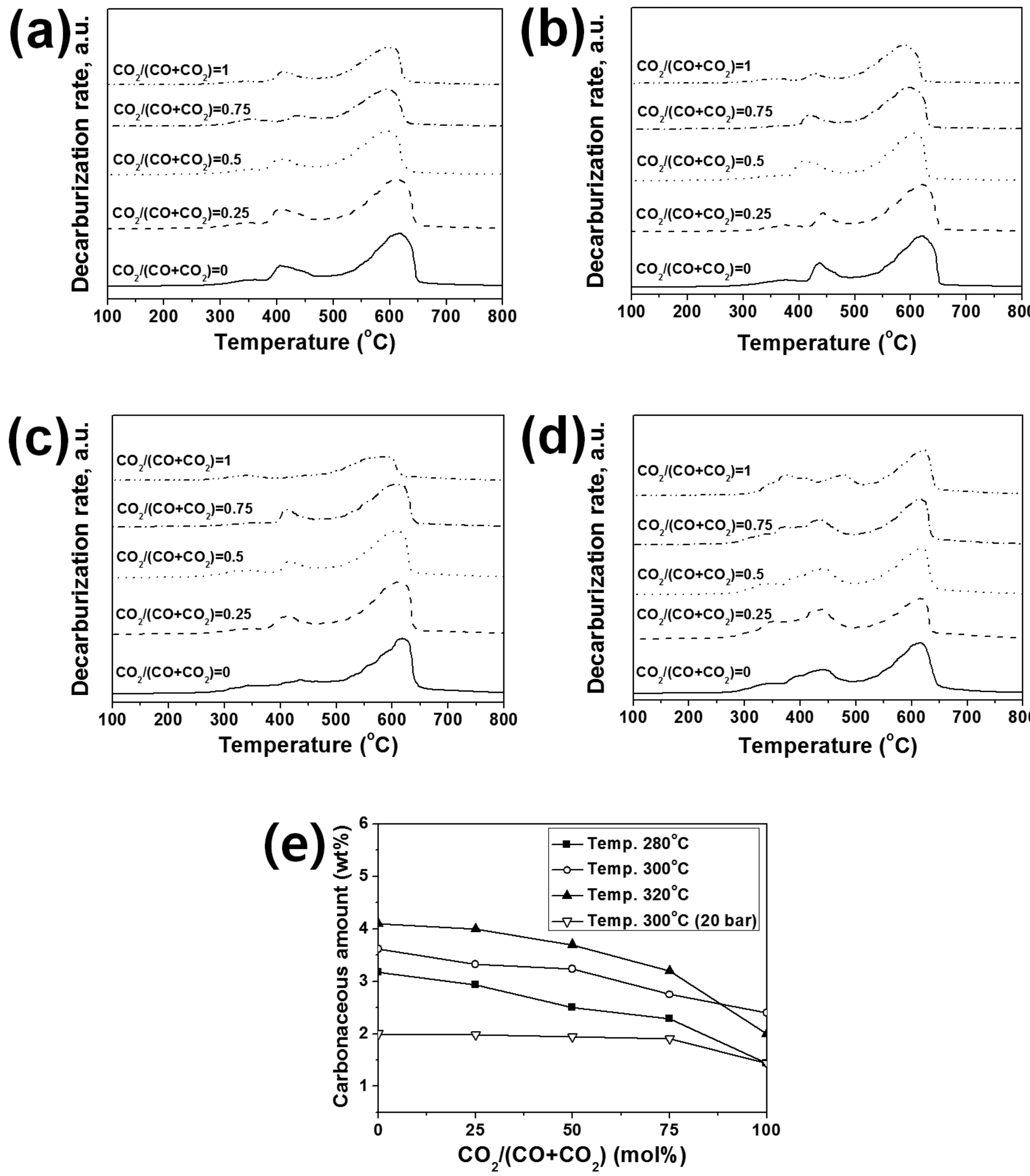

Figure 7. TPDC profiles of the catalysts used at (a) 280 °C and 10 bar, (b) 300 °C and 10 bar, (c) 320 °C and 10 bar, and (d) 300 °C and 20 bar. (e) Carbonaceous material amount calculated from TPDC profiles.

suggested that the presence of CO is more favorable than CO_2 for carbide formation, inducing carbon deposition. Also, Schulz et al.[32] reported that lowering the CO partial pressure of the feed gas would decrease its carbide potential. While the formation of carbide increases with the temperature, it decreases with increasing pressure. The carbidation of Fe in the Fe-catalyzed FTS reaction is known, whereby surface-active carbon formed from CO dissociation is consumed. When the bulk phase of Fe is fully carbided, the carbonaceous materials would be deposited on the surface of catalyst.[41,42] Therefore, the amount of carbon deposition can be inferred from the Fe_xC/Fe_3O_4 ratio in XRD results.

3.4. TPDC Analysis. Panels a–d of Figure 7 show the relative reactivity of the iron carbides that were formed during the FTS reaction at varying $CO_2/(CO + CO_2)$ ratios at each condition. No significant difference among the TPDC profiles of used catalysts could be observed under each set of conditions. The peaks at ∼420 and ∼600 °C in the TPDC profiles were assigned to surface carbides and bulk-phase iron carbides, respectively. The position of the peak maxima shifted to lower temperatures with an increasing $CO_2/(CO + CO_2)$ ratio. Also, the bulk-phase iron carbide peak intensities decreased with an increasing $CO_2/(CO + CO_2)$ ratio. It can be deduced that the stability of the carbides decreases with the CO content in feed gas.[20,32,43,44]

Figure 7e shows the amount of carbide formed during the FTS reaction calculated from integrating peaks in the TPDC profiles. The amount of carbide decreased with an increasing $CO_2/(CO + CO_2)$ ratio and also increased with the reaction temperature. The catalyst used at 300 °C under 20 bar showed much lower carbide formation than those used under 10 bar. It can be deduced that the formation of carbide on the Fe catalyst is favorable at high CO content, high temperature, and low pressure. As explained in the XRD analysis (see section 3.3), the amount of carbon deposition can be inferred from the amount of carbide in Figure 7e. In the case of the reduced

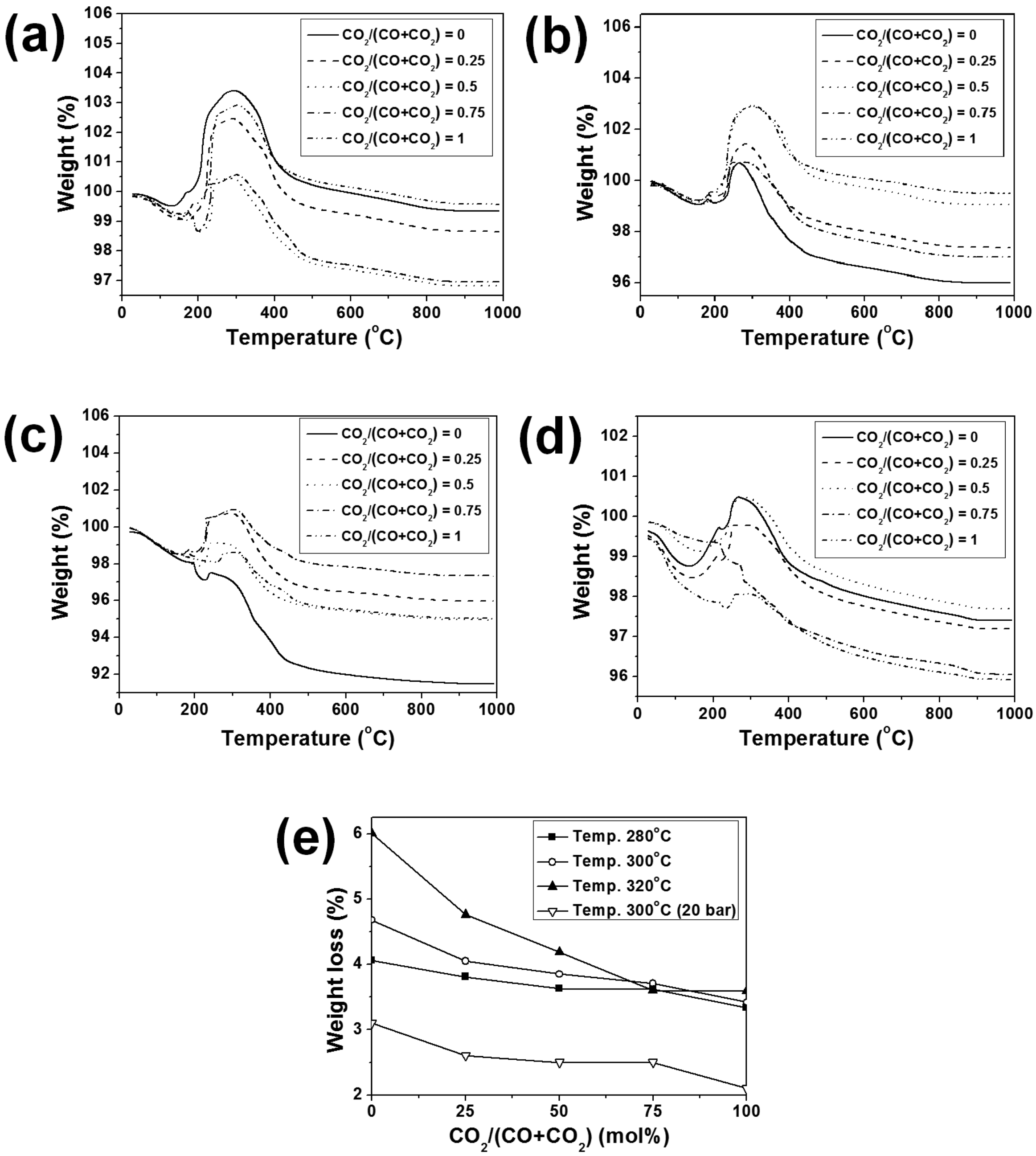

Figure 8. TGA profiles at (a) 280 °C and 10 bar, (b) 300 °C and 10 bar, (c) 320 °C and 10 bar, and (d) 300 °C and 20 bar. (e) Weight loss corresponding to C combustion calculated from TGA profiles.

catalyst, the amount of carbide (3.0%) was higher than some of the used catalyst samples. TGA and elemental analysis showed almost the same results of 3.1 and 3.9%, respectively. It can be inferred that carbide was formed during reduction because of the presence of CO and was then partly converted to HC by high-pressure hydrogenation during the FT reaction.

3.5. TGA. Panels a−d of Figure 8 show the TGA profiles of used catalysts in varying $CO_2/(CO + CO_2)$ ratios under each set of reaction conditions. No significant difference among the used catalyst TGA profiles under each set of conditions could be observed. As shown in panels a−d of Figure 8, after weight loss up to 200 °C by evaporation of moisture absorbed in the catalyst, weight gain as a result of catalyst oxidation was observed at ∼300 °C; subsequent weight loss because of oxidation of deposited HCs that were not removed by washing or carbonaceous materials formed during the FTS reaction was observed with increasing reaction temperature. However, it is difficult to separate the oxidation step of carbonaceous materials from the results of TGA.[45−47]

Figure 8e shows the patterns of weight loss (%) calculated from TGA profiles. The weight losses in the TGA profiles gradually decreased as $CO_2/(CO + CO_2)$ ratios increased, which indicate that a decrease in the CO content in the feed gas results in reduced formation of carbonaceous materials. In addition, the weight losses increased as the reaction temperature increased, which indicate that the increase in the reaction temperature results in greater formation of carbonaceous materials. The catalyst used at 300 °C under 20 bar showed a much lower weight loss than that under 10 bar, which indicates that high pressure is beneficial for reduced formation

Table 1. Elemental Analysis Results of the Catalysts Used under Various Conditions

temperature/pressure	$CO_2/(CO + CO_2)$	C/catalyst (%)	H/catalyst (%)	H/C atomic ratio
280 °C/10 bar	0	5.4	0.5	0.1
	0.25	4.5	0.5	0.1
	0.5	4.1	0.5	0.1
	0.75	3.9	0.3	0.1
	1	2.9	0.5	0.2
300 °C/10 bar	0	5.9	0.6	0.1
	0.25	5.4	0.5	0.1
	0.5	4.7	0.5	0.1
	0.75	4.0	0.4	0.1
	1	3.1	0.2	0.1
320 °C/10 bar	0	6.1	0.7	0.1
	0.25	6.0	0.6	0.1
	0.5	4.9	0.5	0.1
	0.75	4.1	0.4	0.1
	1	3.5	0.3	0.1
300 °C/20 bar	0	3.6	0.4	0.1
	0.25	3.2	0.4	0.1
	0.5	3.2	0.3	0.1
	0.75	3.2	0.4	0.1
	1	2.4	0.2	0.1

of carbonaceous materials. The TGA results were complementary with TPDC results; it can be inferred through TPDC and TGA that the iron carbide may lead to the formation of carbonaceous materials.

3.6. Elemental Analysis. The C and H compositions of used catalyst samples under all reaction conditions were determined by elemental analysis, as seen in Table 1. The H/C ratio for all samples was below 0.2, indicating that the carbon species are mainly composed of carbide or C.[28,40,41] The amount of carbon species decreased with an increasing $CO_2/(CO + CO_2)$ ratio. The carbon content increased with the reaction temperature, which indicates that the carbon deposition increased with the temperature. The catalyst used under 20 bar showed much lower carbon formation than that under 10 bar. The C composition showed almost the same trend as the weight loss (%) in TGA (Figure 8e). On the basis of the above results, it can be suggested that CO is more favorable than CO_2 for the formation of carbide, inducing carbon deposition.

3.7. Carbon Selectivity. Figure 9 shows the patterns of carbon selectivity calculated from the following equation:

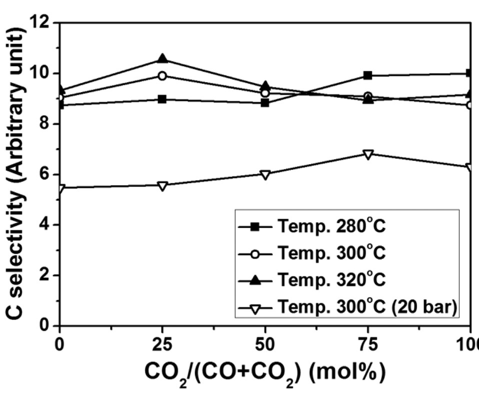

Figure 9. Carbon formation selectivity depending upon $CO_2/(CO + CO_2)$ ratios under various reaction conditions.

$$\text{C selectivity} = \frac{\text{C \% in elemental analysis}}{\text{CO}_x \text{ conversion to HC}} \quad (1)$$

The average C % was calculated from the results of elemental analysis. The carbon selectivity in the present study shows different patterns among the analysis results (TPDC, TGA, and elemental analysis). As determined from analysis results, the amount of carbon deposition over 40 h on-stream was increased with the reaction temperature and CO content in the feed gas. Meanwhile, the carbon selectivity increased with the temperature increases at a low $CO_2/(CO + CO_2)$ ratio; however, selectivity decreased at high $CO_2/(CO + CO_2)$ ratios. In most cases, carbon selectivity increased slightly with a greater $CO_2/(CO + CO_2)$ ratio. In contrast, carbon selectivity decreased with increasing $CO_2/(CO + CO_2)$ ratios at 320 °C under 10 bar; however, there was no significant variation besides the result at 300 °C under 20 bar. Particularly, the carbon selectivity at 300 °C under 20 bar was much lower than those for the reactions under 10 bar, showing the same pattern as compared to the analysis results above. Therefore, it can be deduced that a high pressure is most effective among the reaction condition variables, such as the CO_2 content, temperature, and pressure, for decreasing carbon deposition.

Considering the mechanism of carbon deposition, which is due to full carbidation of the Fe bulk phase on the catalyst surface,[41,42] it can be inferred that the rate of chain propagation is higher than the rate of carbide formation; as a result, the amount of carbon formation is decreased under high-pressure conditions. Although carbon selectivity was not shown to be significantly affected by the CO_2 content, carbon formation over 40 h on-stream decreased with increasing CO_2 content. Consequently, the reaction conditions employing a high $CO_2/(CO + CO_2)$ ratio and pressure are beneficial for preventing carbon deposition in Fe-catalyzed FTS. Additionally, CO_2 consumption by the RWGS reaction was increased with the temperature, even though carbon formation was increased with the temperature. Thus, the FTS using Fe catalysts under high $CO_2/(CO + CO_2)$ ratio, high temperature, and high-pressure conditions corresponds to a process with high carbon efficiency because of high CO_2 utilization.

Many groups have investigated potassium-dependent carbon deposition and have shown that higher potassium content causes an increase in the rate of carbon deposition.[48,49] The optimum potassium content would thus vary depending upon the reaction conditions. Further investigations are therefore necessary to clarify the effect of the potassium content on carbon deposition in connection with the FTS reaction conditions.

4. CONCLUSION

The K/Fe−Cu−Al catalyst showed high performance, irrespective of the $CO_2/(CO + CO_2)$ ratio, in FTS using H_2-sufficient $[H_2/(2CO + 3CO_2) = 1]$ feed gas. The CO_x conversion to HC increased with the temperature and pressure, while it decreased with the $CO_2/(CO + CO_2)$ ratio. Although carbon formation over 40 h on-stream decreased with an increasing $CO_2/(CO + CO_2)$ ratio in the feed gas, there were only small changes in terms of carbon selectivity, irrespective of the reaction conditions, such as the $CO_2/(CO + CO_2)$ ratio and temperature. Carbon selectivity increased slightly at low temperatures as the $CO_2/(CO + CO_2)$ ratio increased, whereas selectivity decreased slightly at high temperatures. However, under high-pressure reaction conditions, carbon formation decreased significantly, as compared to that under low-pressure conditions, indicating that chain propagation is faster than carbide and carbon formation under higher pressures. Consequently, the reaction conditions involving a high $CO_2/(CO + CO_2)$ ratio, high temperature, and high pressure are beneficial for efficient CO_2 conversion and preventing carbon deposition in FTS using Fe catalysts.

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Notes

The authors declare no competing financial interest.

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